Mining Joint Intraimage and Interimage Context for Remote Sensing Change Detection

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***Abstract*— Recent deep learning methods for change detection focus on excavating more discriminative context within individual images. However, due to seasonal change, noise, and so on, the appearance of objects tends to be more heterogeneous among various scenes. Consequently, the above intraimage context is inadequate to represent specific-category objects and pseudo changes would be inevitable in detection results. To deal with this issue, we propose a context aggregation network (CANet) to mine interimage context over all training images for further enhancing intraimage context. Specifically, a Siamese network attached with temporal attention modules is served as a feature encoder to extract multiscale temporal features from bitemporal images. Then, a context extraction module is devised to capture long-range spatial–channel context within individual images. Meanwhile, context representations of underlying categories in the scene are inferred using all training images in an unsupervised manner. Finally, these two kinds of contextual information are aggregated to one which is subsequently fed into a multiscale fusion module to produce the detection map. CANet is compared with several state-of-the-art methods on three benchmark datasets, including the season-varying change detection (SVCD) dataset, the Sun Yat-sen University change detection (SYSU-CD) dataset, and the Learning Vision and Remote Sensing Laboratory building change detection (LEVIR-CD) dataset. It is demonstrated that our method outperforms all comparison methods in terms of F1, overall accuracy (OA), and intersection of union (IoU). The results of CANet on three datasets are available at https://github.com/NuistZF/CANet-for-change-detection and codes will be made public soon.**

***Index Terms*— Change detection, bitemporal remote sensing images, intraimage context, interimage context.**

Manuscript received 11 February 2023; revised 7 April 2023; accepted 4 May 2023. Date of publication 12 May 2023; date of current version 25 May 2023. This work was supported in part by the Natural Science Foundation of China under Grant 62206115, Grant U21B2044, Grant 71972102, and Grant U21B2049; in part by the Key Project for University Natural Science Foundation of Jiangsu Province under Grant 20KJA520002; in part by the Natural Science Foundation of Jiangsu Province under Grant BK20220646; in part by the Humanities and Social Science Fund of Ministry of Education of China under Grant 19YJAZH100; and in part by the Excellent Science and Technology Innovation Team Projects of Jiangsu Province Universities in 2021. *(Corresponding author: Renlong Hang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3275819

I. Introduction

LOCATING changed objects in the remote sensing scene by comparing two images of the same scene taken at different times is the goal of change detection. The detection map could contribute to various applications such as damage assessment [2], urban planning [3], and geological disaster monitoring [4]. At the early stage, pixel-based methods were proposed for change detection [5], [6]. Ordinarily, they conducted pixel-level comparison on bitemporal images to generate the difference image and the detection map was derived via thresholding or clustering algorithms. These methods pay more attention to the features of individual pixels rather than spatial neighborhoods, and thus, they easily produce noisy results [7]. To cope with this issue, several works proposed to divide bitemporal images into homogeneous regions and then capture spatial context from these regions [8], [9]. Nevertheless, they are still based on handcrafted features that encode insufficient variations, failing to work in complex scenes [1].

Recently, deep learning methods have been widely applied and made significant breakthroughs in the remote sensing community [10], [11], [12]. As a common deep learning method, convolutional neural network (CNN) has shown its excellent ability to extract representative features of objects in remote sensing images [13]. Similarly, CNN was also been introduced to compare two patches from the bitemporal images for change detection [14]. Unfortunately, because of the fully connected operation, the vanilla CNN fails to produce the dense prediction, that is to say, the pixel-level prediction. Then, a fully convolutional network (FCN) emerged, which is only composed of convolutional operations. From this point on, FCN and its variants (e.g., UNet) have come to the fore in the pixel-level change detection task [15], [16], [17], [18], [19], [20]. For instance, Daudt et al. [15] developed a UNet-based method named FC-EF that took a composite image consisting of bitemporal images as input to predict the pixel-level detection map. Although UNet-based methods succeed in preserving spatial details by means of skip connections, they have difficulty in capturing multiscale features of objects. To this end, based on UNet, Peng et al. [17] employed dense skip connections to build UNet++ aiming at extracting multiscale features from a combination of bitemporal images. Again, in [20], the difference image of bitemporal images was fed into UNet++ to produce detection results. These methods, which directly fuse bitemporal images via concatenation or differentiation before feeding into the network, are defined as

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"early fusion" methods. Another category of methods known as "late fusion" is supposed to independently extract features from the images at different times by a Siamese architecture and bitemporal features are fused for the inference of detection maps. In [15], two UNet-based "late fusion" methods, including FC-Siam-conc and FC-Siam-diff, were proposed. Both of them replicated the encoder of UNet into two identical ones with parameter sharing and bitemporal images were separately fed into two encoders for feature extraction. The difference between them lies in the decoding phase. FC-Siam-conc concatenated the skip connections from two encoders, while FC-Siam-diff concatenated their absolute difference. Zhang et al. [21] followed the same architecture of FC-Siam-conc and difference discrimination operation was employed in the decoder to get the homogeneous detection map. Above all, these methods focus on studying various connection and fusion strategies to adequately explore context within the spatial neighborhood and make certain achievements. However, they treat both changed information and unchanged information in the context equally without discrimination, limiting the performance of change detection [22], [23].

The attention mechanism emerging in the field of computer vision is meant to help focus on regions of interest in the image [24], [25]. Hence, the attention mechanism becomes the preferred method to enhance changed information and suppress unchanged one. In [26], [27], [28], the convolutional block attention module (CBAM) [25] was used to highlight the features of changed objects in both spatial and channel domains for change detection. In addition to CBAM, DSAMNet [28] employed a deeply supervised module, which could provide supervision for intermediate layers to obtain more robust features. Cheng et al. [1] used channel attention modules, spatial attention modules, and margin maximization modules simultaneously to improve the separability between features of changed and unchanged objects. Fang et al. [29] designed an ensemble channel attention module and it was plugged into UNet++ to refine the features at different semantic levels. In order to capture relations between any two pixels in the image, self-attention was applied for change detection [30], [31], [32]. Because performing self-attention on each pixel is time-consuming, Chen et al. [33] conducted a modified self-attention operation on a few high-level semantic tokens extracted from the image rather than all pixels. Overall, attention-based methods could to some extent improve the detection performance by exploiting the discriminative and long-range context.

Although all the above methods have tried their best to explore more discriminative context within individual images, this intraimage context is struggling to locate the real changed objects under certain circumstances. This is because same-category objects from bitemporal images are likely to exhibit various appearances when encountering seasonal change, noise, and so on. In other words, intraimage context cannot abstract precise semantic information from specific-category objects, further leading to pseudo changes in the detection result. Taking Fig. 1 as an example, the land-cover objects within two marked areas in bitemporal images belong to the same category *vegetation*. However, due to the seasonal change, *vegetation* in two areas $T1$ and $T2$ appears quite different. By directly comparing intraimage context of $T1$ and $T2$, ISNet [1] inevitably produces the pseudo change result X. In order to alleviate the pseudo-change problem, a context aggregation network (CANet), which could aggregate all possible appearance of specific-category objects in the dataset, is proposed to derive interimage context. By integrating interimage context into intraimage one, the pseudo semantic gap between same-category objects could be removed. Therefore, as shown in Fig. 1, when all images associated with *vegetation* are used to generate interimage context, the pseudo change is eliminated in our detection map Y. The main contributions of our work are summarized as follows.

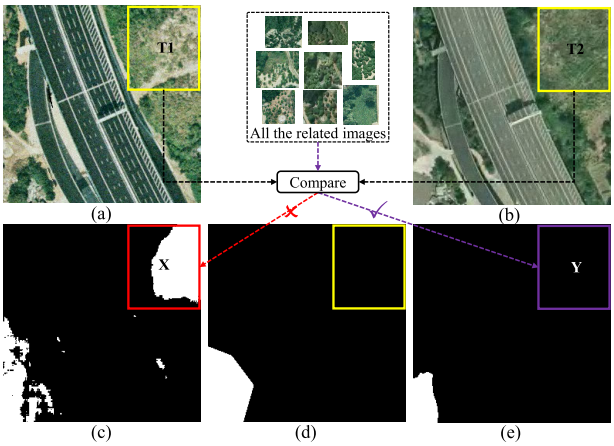


Fig. 1. Leveraging interimage context over all training images for solving the pseudo-change problem. (a) and (b) Bitemporal images. (c) ISNet [1]. (d) Ground truth. (e) CANet. The detection result in the purple box is inferred with the help of interimage context while that in the red box not.

1) A novel CANet is proposed for remote sensing change detection. It adopts a Siamese ResNet18 to extract multiscale features from bitemporal images.
2) We design four modules in CANet to boost the detection performance. The temporal attention module models the temporal dependency between bitemporal features for better locating changed objects. Then, the context memory and extraction modules cooperate to capture hierarchical contextual information for alleviating the pseudo-change problem. Finally, the multiscale fusion module aggregates hierarchical context at each scale for detecting changed objects with various scales.
3) Extensive experiments are conducted on three benchmark datasets to evaluate the performance of CANet. In comparison with several state-of-the-art methods, CANet achieves the best comprehensive accuracy. Furthermore, we perform the ablation study to validate the effectiveness of each module in CANet.

The remainder of this article is organized as follows. Section II elaborates the overall procedure of the proposed CANet. The experimental results on three benchmark datasets and the related analysis are given in Section III. Finally, Section IV concludes this article.

II. Context Aggregation Network

The procedure of CANet is shown in Fig. 2. It includes three main steps: temporal feature extraction, hierarchical context

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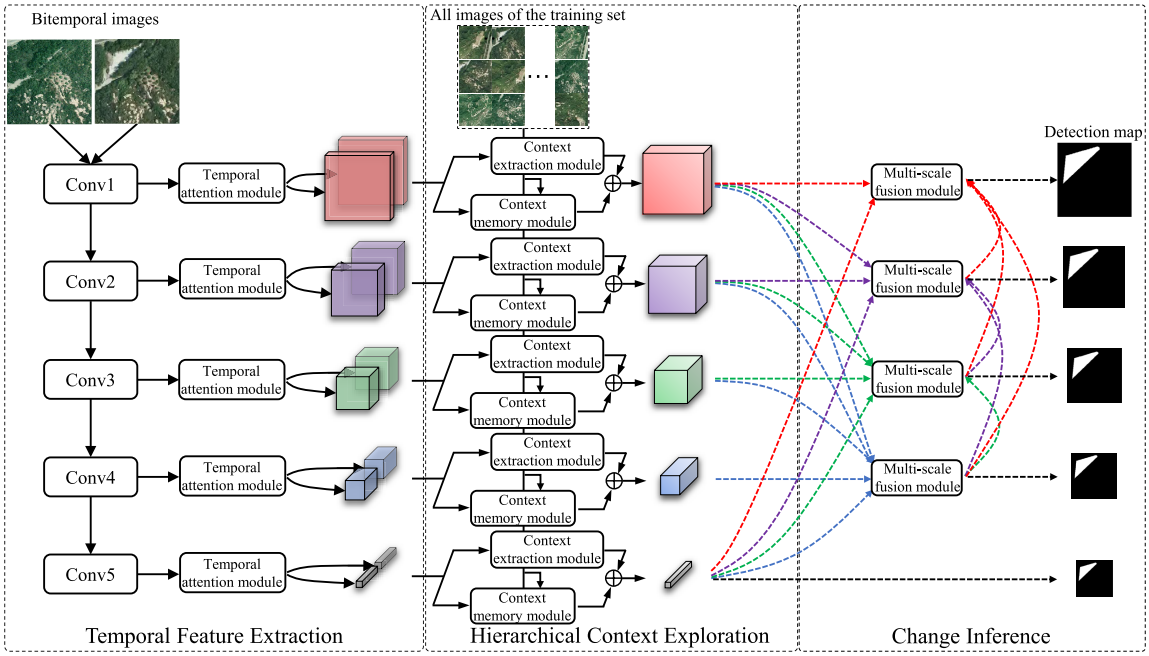


Fig. 2. Overview of our proposed CANet.

exploration, and change inference. In the first step, a Siamese backbone network (ResNet18) is utilized to generate multiscale convolutional features, each of which is subsequently fed into the corresponding temporal attention module for temporal feature extraction. In the second step, a context memory module is employed for mining interimage context over all training images. Meanwhile, the context extraction module is meant to capture the intraimage context from the temporal feature. Then, intraimage and interimage contexts are aggregated to derive a hierarchical context. In the last step, the detection map at each scale is produced by fusing hierarchical context from all scales via a multiscale fusion module. The generated map at the largest scale is selected as the final detection result.

A. Temporal Feature Extraction

How to capture the temporal dependency between bitemporal images would influence the performance of change detection [33]. Most of the existing methods take the strategy of "early" or "late" fusion to directly concatenate bitemporal images/features, ignoring the implied temporal dependency. To remedy this issue, we propose a temporal attention module to explicitly excavate the temporal dependency and encode it into the convolutional feature from the backbone network.

Let us assume that input bitemporal images are defined as $\mathbf{X}_{t1} \in \mathbb{R}^{3\times H\times W}$ and $\mathbf{X}_{t2} \in \mathbb{R}^{3\times H\times W}$, where H and W indicate the height and weight of the input image, respectively. CANet adopts a Siamese network equipped with two identical backbone networks (ResNet18) for extracting multiscale convolutional features by eliminating the fully connected layer of ResNet18. The modified ResNet18 contains five consecutive blocks named Conv1, Conv2, Conv3, Conv4, and Conv5 for abstracting the feature scale by scale. Specifically, Conv1 includes a 7×7 convolutional layer and a 3×3 max-pooling layer, which are both with a stride of 2. After feature extraction, Conv1 reduces the spatial size of $\mathbf{X}_{t1}$ and $\mathbf{X}_{t2}$ to the quarter one $(H/4) \times (W/4)$ and produces two 64-channel output features $\mathbf{F}^1_{t1}$ and $\mathbf{F}^1_{t2}$. The remaining four blocks are all composed of the same group, which includes a 3×3 convolutional layer, a batch normalization layer [34], and ReLU [35]. Note that, the first convolutional layer in Conv3, Conv4, and Conv5 takes a stride of 2, while that of Conv2 takes a stride of 1. Then, Conv2 retains the spatial sizes of input features $\{\mathbf{F}^1_{t1}, \mathbf{F}^1_{t2}\}$ and outputs two 64-channel features. Each of Conv3, Conv4, and Conv5 reduces the spatial size of the input feature by half. The numbers of output channels of these three blocks are 128, 256, and 512. Similarly, the output bitemporal features of last four blocks are represented as $\{\mathbf{F}^2_{t1}, \mathbf{F}^2_{t2}\}$, $\{\mathbf{F}^3_{t1}, \mathbf{F}^3_{t2}\}$, $\{\mathbf{F}^4_{t1}, \mathbf{F}^4_{t2}\}$, and $\{\mathbf{F}^5_{t1}, \mathbf{F}^5_{t2}\}$. To handle the changed objects with various scales, bitemporal features of five blocks are all considered as the inputs of temporal attention modules.

The structure of the temporal attention module is shown in Fig. 3(a). Bitemporal images are essentially a temporal sequence, and hence, convolutional long short-term memory (ConvLSTM) [36], which is known for its ability to model the temporal dependency among the sequence, is selected as a building block. Due to the convolutional recurrent connection, ConvLSTM is capable of modeling the temporal dependency between bitemporal images while preserving spatial information. Specifically, "input" gate $\mathbf{i}_t$, "forget" gate $\mathbf{f}_t$, "output" gate $\mathbf{o}_t$, and candidate cell $\mathbf{C}_t$ constitute ConvLSTM. Each of them adopts a 1×1 convolutional layer with C^s kernels for implementation, where C^s is the number of channels corresponding to the bitemporal features at the sth scale. Given bitemporal features $\{\mathbf{F}^s_{t1}, \mathbf{F}^s_{t2}\} \in \mathbb{R}^{C^s\times H^s\times W^s}$ where $s = 1, 2, 3, 4, 5$, the temporal dependency $\mathbf{H}^s \in \mathbb{R}^{C^s\times H^s\times W^s}$

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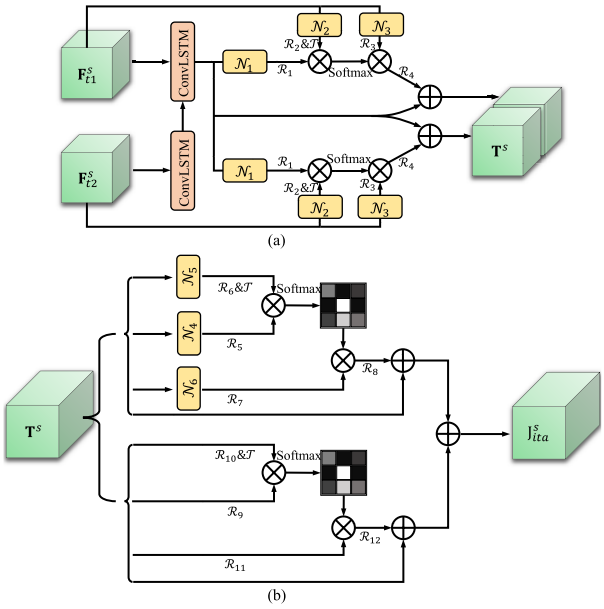


Fig. 3. Structures of (a) temporal attention module and (b) context extraction module. The unit named $\mathcal{N}_\star$ in the figure represents a nonlinear operation consisting of a 3 × 3 convolutional layer, a batch normalization layer, and ReLU. $\mathcal{R}_\star$ denotes the reshape operation. $\mathcal{R}_\star\&\mathcal{T}$ denotes a composite operation, including reshape and transpose operations.

is derived by feeding them into the corresponding temporal attention module in turn as follows:

$$\begin{aligned}
\mathbf{i}_t^s &= \text{sigmoid}\left(\mathbf{W}_{\mathbf{Fi}}^s * \mathbf{F}_t^s + \mathbf{W}_{\mathbf{hi}}^s * \mathbf{h}_{t-1}^s + b_{\mathbf{i}}^s\right)\\
\mathbf{f}_t^s &= \text{sigmoid}\left(\mathbf{W}_{\mathbf{Ff}}^s * \mathbf{F}_t^s + \mathbf{W}_{\mathbf{hf}}^s * \mathbf{h}_{t-1}^s + b_{\mathbf{f}}^s\right)\\
\tilde{C}_t^s &= \tanh\left(\mathbf{W}_{\mathbf{FC}}^s * \mathbf{F}_t^s + \mathbf{W}_{\mathbf{hC}}^s * \mathbf{h}_{t-1}^s + b_{\mathbf{C}}^s\right)\\
\mathbf{C}_t^s &= \mathbf{f}_t^s \circ \mathbf{C}_{t-1}^s + \mathbf{i}_t^s \circ \tilde{C}_t^s\\
\mathbf{o}_t^s &= \text{sigmoid}\left(\mathbf{W}_{\mathbf{Fo}}^s * \mathbf{F}_t^s + \mathbf{W}_{\mathbf{ho}}^s * \mathbf{h}_{t-1}^s + b_{\mathbf{o}}^s\right)\\
\mathbf{h}_t^s &= \mathbf{o}_t^s \circ \tanh\left(\mathbf{C}_t^s\right)
\end{aligned} \tag{1}$$

where "$*$" is the convolutional operation, "$\circ$" is the Hadamard product, and $b_{\mathbf{i}}^s$, $b_{\mathbf{f}}^s$, $b_{\mathbf{C}}^s$, and $b_{\mathbf{o}}^s$ represent the biases. The meanings of weight matrixes can be inferred from their subscripts. For instance, $\mathbf{W}_{\mathbf{Ff}}^s$ denotes the weight between the input feature and "forget" gate. Here, the last output $\mathbf{h}_{t2}^s$ of ConvLSTM is regarded as $\mathbf{H}^s$. Next, we need to encode the temporal dependency $\mathbf{H}^s$ into bitemporal features $\{\mathbf{F}_{t1}^s, \mathbf{F}_{t2}^s\}$. $\mathbf{H}^s$, $\mathbf{F}_{t1}^s$, and $\mathbf{F}_{t2}^s$ are first transformed via a nonlinear operation consisting of a 3 × 3 convolutional layer, a batch normalization layer, and ReLU. Then, we perform the matrix multiplication and softmax function on the transformed results to obtain coefficient matrices $\{\mathbf{A}_{t1}^s, \mathbf{A}_{t2}^s\} \in \mathbb{R}^{H^sW^s \times H^sW^s}$. For each time step, the coefficient matrix is multiplied by the transformed convolutional feature, which is further added to the product to generate the temporal feature. The procedure to generate the temporal feature $\mathbf{T}_{t\in\{t1,t2\}}^s$ using the temporal dependency $\mathbf{H}^s$ and convolutional feature $\mathbf{F}_{t\in\{t1,t2\}}^s$ is formulated as follows:

$$\begin{cases}
\mathbf{A}_t^s = \text{softmax}\left(\mathcal{R}_2\&\mathcal{T}\left(\mathcal{N}_2\left(\mathbf{F}_t^s\right)\right)\mathcal{R}_1\left(\mathcal{N}_1\left(\mathbf{H}^s\right)\right)\right)\\
\mathbf{T}_t^s = \mathcal{R}_4\left(\mathcal{R}_3\left(\mathcal{N}_3\left(\mathbf{F}_t^s\right)\right)\mathbf{A}_t^s\right) + \mathbf{H}^s
\end{cases} \tag{2}$$

where $\mathcal{N}_\star(\cdot)$ represents the aforementioned nonlinear operation, $\mathcal{R}_\star(\cdot)$ denotes the reshape operation, and $\mathcal{R}_\star\&\mathcal{T}(\cdot)$ denotes a composite operation, including reshape and transpose operations. In the following, these three operations will be used again and descriptions on them are no longer given. By concatenating temporal features at two time points, the final temporal feature $\mathbf{T}^s \in \mathbb{R}^{2C^s \times H^s \times W^s} = [\mathbf{T}_{t1}^s; \mathbf{T}_{t2}^s]$ is obtained.

B. Hierarchical Context Exploration

The goal of this step is to simultaneously capture intraimage and interimage contexts for locating the real changed objects in the remote sensing scene. In the following, we will elaborate how to explore these two kinds of contextual information.

It has been demonstrated that long-range contextual information is vital to understand the scene [37]. Recently, many works, such as BIT [33], designed various self-attention-based modules to capture long-range spatial context, but they ignore long-range dependency existing in the channel domain. As is known to all, each channel map corresponds to the response of a specific category and relationships between channel maps could help highlight the feature of a specific category. Hence, motivated in [37], we introduce a dual attention mechanism shown in Fig. 3(b) to capture long-range spatial–channel context. Given the temporal feature $\mathbf{T}^s$, self-attention is applied in the spatial domain to generate the long-range spatial context $\mathbf{J}_{sp}^s \in \mathbb{R}^{2C^s \times H^s \times W^s}$ as follows:

$$\begin{cases}
\mathbf{A}_{sp}^s = \text{softmax}\left(\mathcal{R}_6\&\mathcal{T}\left(\mathcal{N}_5\left(\mathbf{T}^s\right)\right)\mathcal{R}_5\left(\mathcal{N}_4\left(\mathbf{T}^s\right)\right)\right)\\
\mathbf{J}_{sp}^s = \mathcal{R}_8\left(\mathcal{R}_7\left(\mathcal{N}_6\left(\mathbf{T}^s\right)\right)\mathbf{A}_{sp}^s\right) + \mathbf{T}^s
\end{cases} \tag{3}$$

where the spatial attention map $\mathbf{A}_{sp}^s \in \mathbb{R}^{H^sW^s \times H^sW^s}$ reflects the similarity between any two pixels in $\mathbf{T}^s$. For the specific position in $\mathbf{J}_{sp}^s$, it is the combination of the features from all other positions, that is to say, $\mathbf{J}_{sp}^s$ successfully aggregates long-range spatial context. With respect to the channel domain, we adopt a similar strategy to capture long-range context. The difference is that there is no need to perform the nonlinear operation and the channel attention map $\mathbf{A}_{ch}^s \in \mathbb{R}^{2C^s \times 2C^s}$ could be directly calculated from $\mathbf{T}^s$. The long-range channel context $\mathbf{J}_{ch}^s \in \mathbb{R}^{2C^s \times H^s \times W^s}$ is formulated as follows:

$$\begin{cases}
\mathbf{A}_{ch}^s = \text{softmax}\left(\mathcal{R}_9\left(\mathbf{T}^s\right)\mathcal{R}_{10}\&\mathcal{T}\left(\mathbf{T}^s\right)\right)\\
\mathbf{J}_{ch}^s = \mathcal{R}_{12}\left(\mathbf{A}_{ch}^s\mathcal{R}_{11}\left(\mathbf{T}^s\right)\right) + \mathbf{T}^s.
\end{cases} \tag{4}$$

Similar to $\mathbf{J}_{sp}^s$, $\mathbf{J}_{ch}^s$ stores long-range channel contextual information because it aggregates the features over all channel maps. $\mathbf{J}_{sp}^s$ and $\mathbf{J}_{ch}^s$ are fused via elementwise summation to obtain the final long-range spatial–channel context $\mathbf{J}_{\text{ita}}^s \in \mathbb{R}^{2C^s \times H^s \times W^s}$.

The context extraction module aims at capturing rich contextual information within individual images. However, seasonal change and other external factors might make the appearance of same-category objects quite different among scenes. Thus, individual images could barely provide precise enough semantic information for the specific category. As a result, this intraimage context would mistakenly detect same-category objects with different appearances at two time points as changed objects. If we could collect all possible appearances (context) of objects with the same category over the

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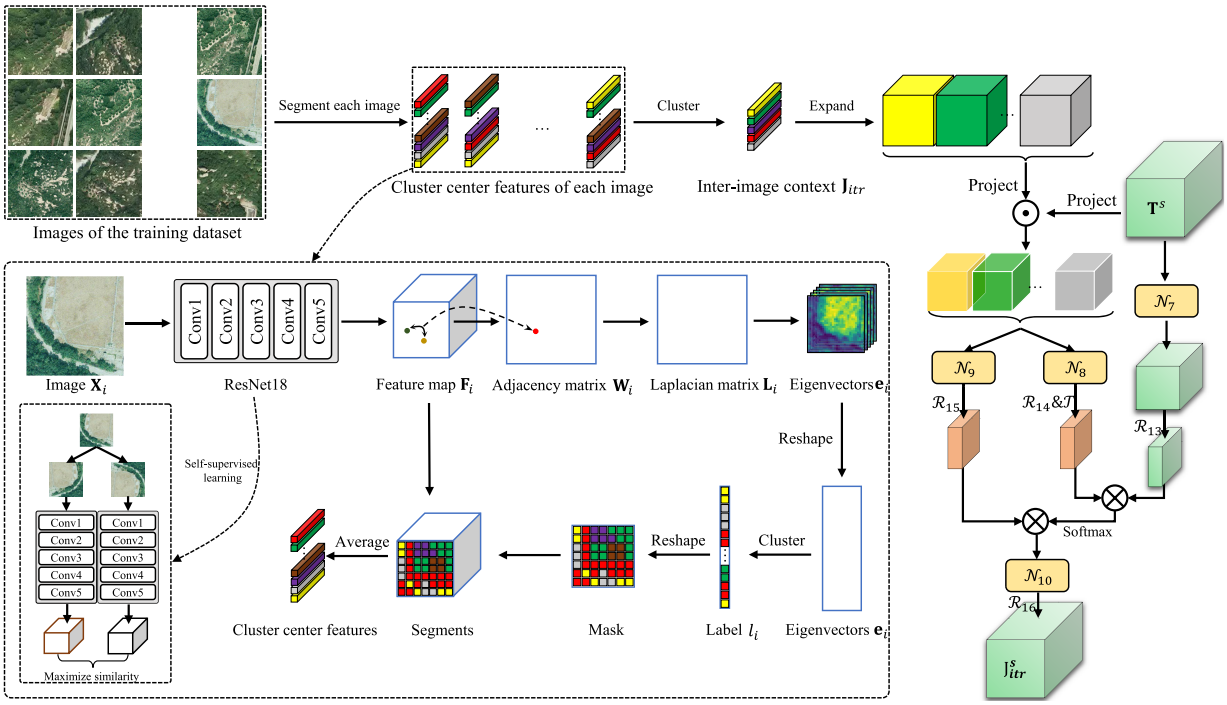


Fig. 4. Structure of context memory module. The unit named $\mathcal{N}_\star$ in the figure represents a nonlinear operation consisting of a 3 × 3 convolutional layer, a batch normalization layer, and ReLU. $\mathcal{R}_\star$ denotes the reshape operation. $\mathcal{R}_\star\&\mathcal{T}$ denotes a composite operation, including reshape and transpose operations.

dataset, the pseudo-change problem would be eased. To this end, we design a context memory module shown in Fig. 4 to explore this interimage context over all training images.

As long as regions of each category in each training image are known to us in advance, then we could aggregate region features of same categories to produce the interimage context. However, such ground truth of each training image is unavailable in the change detection task. This demands us to divide each training image into multiple homogeneous regions in an unsupervised manner. Accordingly, we consider using spectral clustering to group each pixel to the corresponding category. Traditional spectral clustering [38] constructs the graph based on shallow features such as color information, limiting the discrimination capacity. Therefore, we propose to conduct spectral clustering on deep convolutional features for improving the performance.

Specifically, similar to [39], two views of each input image are generated based on data augmentation such as flipping and rotation, and they are converted into convolutional features using ResNet18. The goal of training ResNet18 is to maximize the similarity between the features related to two views of the same input image and minimize that between the features related to any views of different input images. Let $\mathbf{X}_i \in \mathbb{R}^{3\times H\times W}$ be the input image. When ResNet18 is well trained, $\mathbf{X}_i$ could be converted to the corresponding convolutional feature $\mathbf{F}_i \in \mathbb{R}^{128\times(H/8)\times(W/8)}$. Next, we need to regard the reshaped convolutional feature $\mathbf{F}_i \in \mathbb{R}^{128\times(HW/64)}$ as an undirected graph $G_i = (V_i, E_i)$, where $V_i = \{\mathbf{F}_i[h] \in \mathbb{R}^{128} | 1 \le h \le (HW/64)\}$ is the set of all pixels in $\mathbf{F}_i$ and E_i is the set of all edges joining any two pixels. Then, the adjacency matrix $\mathbf{W}_i \in \mathbb{R}^{(HW/64)\times(HW/64)}$ could be inferred based on radial basis function as follows:

$$\mathbf{W}_i[h, w] = e^{\frac{-\|\mathbf{F}_i[h]-\mathbf{F}_i[w]\|^2}{2\sigma^2}} \tag{5}$$

where σ is set to 2.5. After obtaining $\mathbf{W}_i$, graph cut is used to divide the original graph G_i into K subgraphs $\{R_i^k | 1 \le k \le K, R_i^1 \bigcap \cdots R_i^k \cdots \bigcap R_i^K = \emptyset, R_i^1 \bigcup \cdots R_i^k \cdots \bigcup R_i^K = G_i\}$ by minimizing the following equation:

$$\text{cut}\left(R_i^1, \ldots, R_i^K\right) = \sum_{k=1}^{K} \sum_{h\in R_i^k, w\in \overline{R_i^k}} \mathbf{W}_i[h, w] \tag{6}$$

where $\overline{R_i^k}$ represents the complementary set of R_i^k.

For implementing graph cut, the normalized Laplacian matrix $\mathbf{L}_i \in \mathbb{R}^{(HW/64)\times(HW/64)}$ is used to derive the eigenvectors that are subsequently clustered to produce image segments. First, we need to calculate the degree matrix $\mathbf{D}_i \in \mathbb{R}^{(HW/64)\times(HW/64)}$ of $\mathbf{W}_i$. $\mathbf{D}_i$ is a diagonal matrix where $\mathbf{D}_i[h, h] = \sum_w \mathbf{W}_i[h, w]$. Based on $\mathbf{W}_i$ and $\mathbf{D}_i$, $\mathbf{L}_i$ is built as follows:

$$\mathbf{L}_i = \mathbf{D}_i^{-\frac{1}{2}}(\mathbf{D}_i - \mathbf{W}_i)\mathbf{D}_i^{-\frac{1}{2}}. \tag{7}$$

Here, the eigenvectors of $\mathbf{L}_i$ corresponding to the first four smallest eigenvalues are selected and concatenated. After performing k-means clustering on the concatenated eigenvectors $\mathbf{e}_i \in \mathbb{R}^{4\times(HW/64)}$, the label $l_i \in \{1, \ldots, K\}$, where K is set to 10, of each pixel in $\mathbf{F}_i$ is assigned. For each input image $\mathbf{X}_i$, the average feature of the segments belonging to the same cluster is calculated. Finally, we collect these average features from all training images and perform k-means clustering on them. Different from the first k-means clustering, this one

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sets the number N of clusters to the number of categories of changed objects. The cluster center features generated from the second k-means clustering are roughly regarded as the representations of specific-category changed objects. Thus, the desired interimage context $\mathbf{J}_{\text{itr}} \in \mathbb{R}^{128\times N}$ is the concatenation of all cluster center features.

The last part of the context memory module is to encode the interimage context $\mathbf{J}_{\text{itr}}$ into the temporal feature $\mathbf{T}^s$. Now, we take the interimage context $\mathbf{J}_{\text{itr}}[n] \in \mathbb{R}^{128}$ related to changed objects of the nth category as an example to illustrate this encoding procedure. $\mathbf{J}_{\text{itr}}[n]$ is first expanded to $128 \times H^s \times W^s$ by replicating itself along with the spatial dimension. Next, $\mathbf{J}_{\text{itr}}[n]$ and $\mathbf{T}^s$ are projected to the same feature space with dimension of $(C^s/2)\times H^s \times W^s$ via a linear operation. The Hadamard product and sigmoid function are performed on these two features to generate the intermediate feature $\mathbf{I}^s \in \mathbb{R}^{(C^s/2)\times H^s\times W^s}$ as follows:

$$\mathbf{I}^s = \mathcal{L}\big(\text{sigmoid}\big(\mathcal{L}(\mathbf{J}_{\text{itr}}[n]) \circ \mathcal{L}\big(\mathbf{T}^s\big)\big)\big) \tag{8}$$

where $\mathcal{L}(\cdot)$ is a linear operation consisting of a fully connected layer with $(C^s/2)$ hidden nodes. Then, we measure the similarity matrix $\mathbf{A}^{s,n} \in \mathbb{R}^{H^sW^s\times H^sW^s}$ between $\mathbf{T}^s$ and $\mathbf{I}^s$, and $\mathbf{I}^s$ is enhanced by multiplying itself with $\mathbf{A}^{s,n}$

$$\begin{cases} \mathbf{A}^{s,n} = \text{softmax}\big(\mathcal{R}_{14}\&\mathcal{T}\big(\mathcal{N}_8\big(\mathbf{I}^s\big)\big)\mathcal{R}_{13}\big(\mathcal{N}_7\big(\mathbf{T}^s\big)\big)\big) \\ \bar{J}^n_{\text{itr}} = \mathcal{R}_{16}\big(\mathcal{R}_{15}\big(\mathcal{N}_9\big(\mathbf{I}^s\big)\big)\mathbf{A}^{s,n}\big). \end{cases} \tag{9}$$

$\bar{J}^n_{\text{itr}} \in \mathbb{R}^{(C^s/2)\times H^s\times W^s}$ encodes the interimage context of the nth category into the temporal feature. When we get N enhanced temporal features, they are concatenated along with the channel dimension and a nonlinear operation is further performed on the concatenation result to obtain the final enhanced feature $\mathbf{J}^s_{\text{itr}} \in \mathbb{R}^{2C^s\times H^s\times W^s} = \mathcal{N}_{10}([\bar{J}^1_{\text{itr}};\ldots;\bar{J}^N_{\text{itr}}])$. Ultimately, $\mathbf{J}^s_{\text{ita}}$ and $\mathbf{J}^s_{\text{itr}}$ are fused via elementwise summation to obtain the final encoding feature $\mathbf{F}^s_e$. $\mathbf{F}^s_e$ leverages the intraimage and interimage contexts simultaneously to provide more precise semantic information for changed objects.

C. Change Inference

Inspired by [40], we introduce a multiscale fusion module to aggregate $\mathbf{F}^s_e$ at all scales to simultaneously take full advantage of low-level (e.g., spatial details) and high-level (e.g., semantic information) information. The decoding feature $\mathbf{F}^s_d$ used for inferring the detection map is generated by fusing multiscale $\mathbf{F}^s_e$. In the following, we take $\mathbf{F}^3_d$ as an example to illustrate the whole procedure shown in Fig. 5.

Given two low-level encoding features $\mathbf{F}^1_e$ and $\mathbf{F}^2_e$, they are first downsampled to the same spatial size of $\mathbf{F}^3_e$. Specifically, a 2×2 max-pooling layer with the stride of 2 is applied to $\mathbf{F}^1_e$ and $\mathbf{F}^2_e$ for downsampling. Then, the decoding operation $\mathcal{H}(\cdot)$ comprised by a 3×3 convolutional layer with 64 kernels, a batch normalization layer, and ReLU is performed on the downsampled $\mathbf{F}^1_e$ as well as $\mathbf{F}^2_e$. Meanwhile, two high-level decoding features $\mathbf{F}^4_d$ and $\mathbf{F}^5_d$ containing semantic information are upsampled to the spatial size of $\mathbf{F}^3_e$ via $\times 2$ and $\times 4$ bilinear interpolation, respectively. Similarly, upsampled $\mathbf{F}^4_d$ and $\mathbf{F}^5_d$ are fed into $\mathcal{H}$. With respect to $\mathbf{F}^3_e$, it is directly transformed by the decoding operation $\mathcal{H}$. The feature aggregation operation

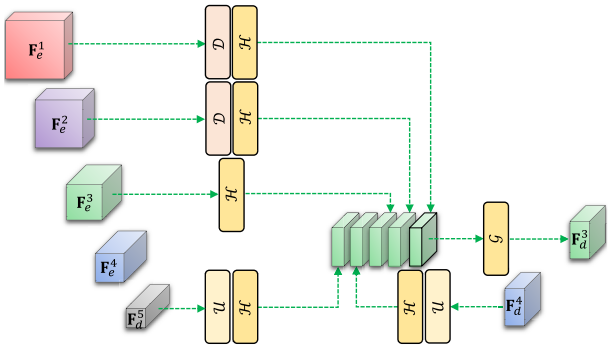


Fig. 5. Structure of multiscale fusion module. $\mathcal{D}$ and $\mathcal{U}$ represent downsampling and upsampling operations, respectively. $\mathcal{G}$ consists of a 3×3 convolutional layer with 320 kernels, a batch normalization layer, and ReLU. $\mathcal{H}$ is comprised of a 3×3 convolutional layer with 64 kernels, a batch normalization layer, and ReLU. Note that, the multiscale fusion module is not performed on $\mathbf{F}^5_e$, and thus, $\mathbf{F}^5_e$ could be directly regarded as $\mathbf{F}^5_d$.

$\mathcal{G}(\cdot)$ is further conducted on the concatenation of the above five same-size features to produce the decoding feature $\mathbf{F}^3_d$. $\mathcal{G}$ consists of a 3×3 convolutional layer with 320 kernels, a batch normalization layer, and ReLU. Generally, the decoding feature $\mathbf{F}^s_d$ at the sth scale is computed as follows:

$$\mathbf{F}^s_d = \begin{cases} \mathbf{F}^s_e & s = 5 \\ \mathcal{G}\Big(\Big[\mathcal{H}\big(\mathcal{D}\big(\mathbf{F}^z_e\big)\big)^{s-1}_{z=1}; \mathcal{H}\big(\mathbf{F}^s_e\big); \mathcal{H}\big(\mathcal{U}\big(\mathbf{F}^z_d\big)\big)^{5}_{z=s+1}\Big]\Big) & 1\le s\le 4 \end{cases} \tag{10}$$

where $\mathcal{D}(\cdot)$ and $\mathcal{U}(\cdot)$ represent downsampling and upsampling operations, respectively.

Decoding features at five scales are separately transformed to the corresponding detection map via a 3×3 convolutional layer with two kernels. Finally, multiscale detection maps are resized to the spatial size $H\times W$ via bilinear interpolation. We choose cross entropy as the loss function for CANet. Let $\mathbf{P}^s \in \mathbb{R}^{H\times W}$ be the predicted detection map at the sth scale. $\mathbf{Y} \in \mathbb{R}^{H\times W}$ is the ground truth where 1 indicates the change and 0 indicates the nonchange. The final loss function is formulated as follows:

$$L = -\frac{1}{HW}\sum_{s=1}^{5}\sum_{h=1}^{H}\sum_{w=1}^{W}\big[\mathbf{Y}_{h,w}\log\mathbf{P}^s_{h,w} + \big(1-\mathbf{Y}_{h,w}\big)\log\big(1-\mathbf{P}^s_{h,w}\big)\big]. \tag{11}$$

III. Experimental Results

Three public datasets, including the season-varying change detection (SVCD) dataset [41], the Learning Vision and Remote Sensing Laboratory building change detection (LEVIR-CD) dataset [30], and the Sun Yat-sen University change detection (SYSU-CD) dataset [28], are chosen for validating the effectiveness of the proposed CANet. In the following, three datasets and implementation details are first introduced. Then, experimental results of comparison methods on three datasets are reported and analyzed. Finally, extensive ablation studies are conducted to verify the significance of each component in CANet.

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TABLE I
CONFIGURATION OF CONVOLUTIONAL KERNELS IN NONLINEAR OPERATIONS

Layer \ Scale	$s=1$	$s=2$	$s=3$	$s=4$	$s=5$
$\mathcal{N}_1$	$3\times3\times8$	$3\times3\times8$	$3\times3\times16$	$3\times3\times32$	$3\times3\times64$
$\mathcal{N}_2$	$3\times3\times8$	$3\times3\times8$	$3\times3\times16$	$3\times3\times32$	$3\times3\times64$
$\mathcal{N}_3$	$3\times3\times64$	$3\times3\times64$	$3\times3\times128$	$3\times3\times256$	$3\times3\times512$
$\mathcal{N}_4$	$1\times1\times16$	$1\times1\times16$	$1\times1\times32$	$1\times1\times64$	$1\times1\times128$
$\mathcal{N}_5$	$1\times1\times16$	$1\times1\times16$	$1\times1\times32$	$1\times1\times64$	$1\times1\times128$
$\mathcal{N}_6$	$1\times1\times128$	$1\times1\times128$	$1\times1\times256$	$1\times1\times512$	$1\times1\times1024$
$\mathcal{N}_7$	$1\times1\times32$	$1\times1\times32$	$1\times1\times64$	$1\times1\times128$	$1\times1\times256$
$\mathcal{N}_8$	$1\times1\times32$	$1\times1\times32$	$1\times1\times64$	$1\times1\times128$	$1\times1\times256$
$\mathcal{N}_9$	$1\times1\times32$	$1\times1\times32$	$1\times1\times64$	$1\times1\times128$	$1\times1\times256$
$\mathcal{N}_{10}$	$1\times1\times128$	$1\times1\times128$	$1\times1\times256$	$1\times1\times512$	$1\times1\times1024$

TABLE II
DIMENSION CHANGE OF FEATURES AFTER RESHAPE OPERATIONS

	Before	Later
$\mathcal{R}_1$	$\frac{C^s}{8}\times H^s\times W^s$	$\frac{C^s}{8}\times H^sW^s$
$\mathcal{R}_2$	$\frac{C^s}{8}\times H^s\times W^s$	$\frac{C^s}{8}\times H^sW^s$
$\mathcal{R}_3$	$C^s\times H^s\times W^s$	$C^s\times H^sW^s$
$\mathcal{R}_4$	$C^s\times H^sW^s$	$C^s\times H^s\times W^s$
$\mathcal{R}_5$	$\frac{C^s}{4}\times H^s\times W^s$	$\frac{C^s}{4}\times H^sW^s$
$\mathcal{R}_6$	$\frac{C^s}{4}\times H^s\times W^s$	$\frac{C^s}{4}\times H^sW^s$
$\mathcal{R}_7$	$2C^s\times H^s\times W^s$	$2C^s\times H^sW^s$
$\mathcal{R}_8$	$2C^s\times H^sW^s$	$2C^s\times H^s\times W^s$
$\mathcal{R}_9$	$2C^s\times H^s\times W^s$	$2C^s\times H^sW^s$
$\mathcal{R}_{10}$	$2C^s\times H^s\times W^s$	$2C^s\times H^sW^s$
$\mathcal{R}_{11}$	$2C^s\times H^s\times W^s$	$2C^s\times H^sW^s$
$\mathcal{R}_{12}$	$2C^s\times H^sW^s$	$2C^s\times H^s\times W^s$
$\mathcal{R}_{13}$	$\frac{C^s}{2}\times H^s\times W^s$	$\frac{C^s}{2}\times H^sW^s$
$\mathcal{R}_{14}$	$\frac{C^s}{2}\times H^s\times W^s$	$\frac{C^s}{2}\times H^sW^s$
$\mathcal{R}_{15}$	$\frac{C^s}{2}\times H^s\times W^s$	$\frac{C^s}{2}\times H^sW^s$
$\mathcal{R}_{16}$	$\frac{C^s}{2}\times H^sW^s$	$\frac{C^s}{2}\times H^s\times W^s$

A. Datasets

They are listed in the following.

1) *SVCD:* It is originally comprised of seven pairs of the season-varying image with the resolution of 0.03–1 m taken by Google Earth. According to [41], they are cropped into 15 998 pairs of bitemporal images with the spatial size of 256 × 256. Among all bitemporal images, 10 000, 2998, and 3000 pairs are separately selected for training, validation, and testing. Car change, tank change, land change, building change, and road change are included in change types.
2) *LEVIR-CD:* The 637 image pairs with the spatial size of 1024 × 1024 in LEVIR-CD were acquired by Google Earth between 2002 and 2018 in Texas, USA. The spatial resolution of them is 0.5 m. According to [30], 10 192 pairs of 256 × 256 bitemporal images are obtained by cropping original images. Then, the numbers of training, validation, and testing sets are set to 7120, 1024, and 2048, respectively. LEVIR-CD only concerns two change types, including newly constructed buildings and destructed buildings.
3) *SYSU-CD:* The 20 000 pairs of 0.5-m resolution bitemporal images with the spatial size of 256 × 256 constitute SYSU-CD. They were acquired between 2007 and 2014 in Hong Kong. All bitemporal images are further divided into the training set with 12 000 pairs, the validation set with 4000 pairs, and the testing set with 4000 pairs. The change types in SYSU-CD include newly built urban buildings, suburban dilation, groundwork before construction, change of vegetation, road expansion, and sea construction.

B. Implementation Details

All the methods, including CANet and other comparison methods, are implemented with PyTorch [42] on a workstation equipped with an Intel i7-10700 CPU and a single NVIDIA Geforce RTX 3090 GPU. AdamW [43] is chosen to optimize the proposed CANet. Specifically, β_1, β_2, and ϵ are set to 0.9, 0.999, and 10^{-8}, respectively. The initial learning rate is set to 0.001. The number of training epochs is 200 and the batch size is 8. After each training epoch, we perform validation and the best model is used to obtain the detection results of testing set. For each dataset, ResNet18 used in CANet is pretrained on the corresponding training set via SimCLR [39]. In addition, spectral clustering is performed to produce interimage contextual information before training CANet. According to the description in Section II-B, N is equal to the number of change types. More concretely, it is set to 5, 2, and 6 for SVCD, LEVIR-CD, and SYSU-CD, respectively. The configuration of all nonlinear operations $\mathcal{N}_\star$ is listed in Table I and the dimension change of features after reshape operations is shown in Table II. To evaluate the performance of change detection methods, we choose five metrics, including precision, recall, F1 score, overall accuracy (OA), and intersection of union (IoU). All of them can be calculated as follows:

$$\begin{aligned}\text{Precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}}\\ \text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}}\\ \text{F1} &= \frac{2\text{TP}}{2\text{TP}+\text{FP}+\text{FN}}\\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}}\\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}}\end{aligned} \tag{12}$$

where TP and TN represent the numbers of changed and unchanged pixels that are correctly classified, respectively, and FP and FN indicate the numbers of changed and unchanged pixels that are wrongly classified.

C. Comparison With State-of-the-Art Methods

We select seven state-of-the-art change detection methods, including FC-EF [15], FC-Siam-conc [15], FC-Siam-diff [15], SNUNet [29], DSAMNet [28], BIT [33], and ISNet [1] for comparison. These methods are implemented with default hyperparameters according to their public codes.

The experimental results on SVCD are listed in Table III. It can be found that three UNet-based methods (FC-EF,

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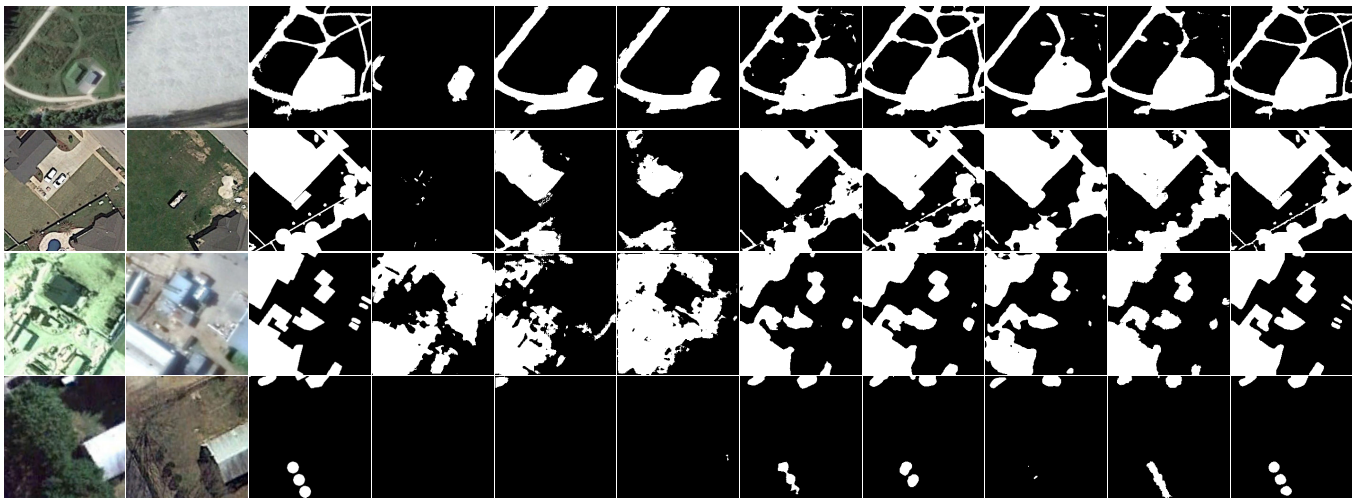


Fig. 6. Detection maps of comparison methods on SVCD. (a) Image at $t1$. (b) Image at $t2$. (c) Ground truth. (d) FC-EF [15]. (e) FC-Siam-conc [15]. (f) FC-Siam-diff [15]. (g) SNUNet [29]. (h) DSAMNet [28]. (i) BIT [33]. (j) ISNet [1]. (k) CANet.

TABLE III
EXPERIMENTAL RESULTS ON SVCD

Method	Precision	Recall	F1	OA	IoU
FC-EF [15]	81.81	41.08	54.70	91.97	37.64
FC-Siam-conc [15]	84.15	54.88	66.44	93.46	49.74
FC-Siam-diff [15]	85.69	44.87	58.90	92.61	41.74
SNUNet [29]	95.19	92.51	93.83	98.56	88.38
DSAMNet [28]	90.24	**97.43**	93.70	98.45	88.14
BIT [33]	94.57	88.18	91.26	98.01	83.93
ISNet [1]	95.18	94.43	94.80	98.78	90.12
CANet	**96.82**	97.29	**97.05**	**99.30**	**94.27**

TABLE IV
EXPERIMENTAL RESULTS ON LEVIR-CD

Method	Precision	Recall	F1	OA	IoU
FC-EF [15]	84.70	77.18	80.76	98.13	67.73
FC-Siam-conc [15]	87.32	81.28	84.19	98.45	72.70
FC-Siam-diff [15]	88.46	81.10	84.62	98.50	73.34
SNUNet [29]	91.38	89.05	90.20	99.01	82.14
DSAMNet [28]	77.82	**92.76**	84.63	98.28	73.36
BIT [33]	90.34	88.31	89.31	98.92	80.69
ISNet [1]	92.46	88.27	90.32	99.04	82.35
CANet	**92.52**	90.06	**91.27**	**99.12**	**83.95**

TABLE V
EXPERIMENTAL RESULTS ON SYSU-CD

Method	Precision	Recall	F1	OA	IoU
FC-EF [15]	80.37	66.77	72.94	88.32	57.41
FC-Siam-conc [15]	81.75	66.58	73.39	88.61	57.96
FC-Siam-diff [15]	**89.41**	51.08	65.02	87.04	48.17
SNUNet [29]	74.82	77.22	76.00	88.50	61.29
DSAMNet [28]	72.90	**84.16**	78.12	88.89	64.10
BIT [33]	80.54	75.36	77.86	89.90	63.75
ISNet [1]	80.27	76.41	78.29	90.01	64.44
CANet	82.80	81.22	**82.00**	**91.59**	**69.50**

FC-Siam-conc, and FC-Siam-diff) obtain the worst performance in terms of all metrics. By introducing the attention mechanism to highlight the features of changed objects, the remaining comparison methods improve the detection results remarkably. Among them, ISNet achieves the best performance due to the margin maximization strategy, which aims at refining the boundary of changed objects. Compared with ISNet, the proposed CANet improves F1, OA, and IoU to 97.05% (2.25 ↑), 99.30% (0.52 ↑), and 94.27% (4.15 ↑), respectively. This significant improvement is mainly due to the introduction of hierarchical contextual information obtained by fusing intraimage and interimage contexts. Fig. 6 shows the detection maps of all methods on the testing set of SVCD. It can be observed that UNet-based methods only detect a few changed objects and thus generate the terrible visual results. Due to the attention mechanism, SNUNet, DSAMNet, BIT, and ISNet manage to roughly locate all changed objects. However, these methods inevitably encounter the pseudo-change problem. Consequently, the detected changed regions are still incomplete and inaccurate. When the pseudo-change problem is addressed by CANet, the boundaries of detected changed objects are more sharp and the intraobject semantic consistence is better.

The similar conclusions can be drawn from LEVIR-CD and SYSU-CD. Tables IV and V list the quantitative results on these two datasets and CANet still obtains the best detection performance. Specifically, compared with the second-best method, CANet improves F1, OA, and IoU to 91.27% (0.95 ↑), 99.12% (0.08 ↑), and 83.95% (1.60 ↑), respectively, on LEVIR-CD. With respect to SYSU-CD, F1, OA, and IoU are separately improved to 82.00% (3.71 ↑), 91.59% (1.58 ↑), and 69.50% (5.06 ↑), respectively. As shown in Figs. 7 and 8, CANet generates much more satisfactory visual results, in comparison with other methods.

In addition to the detection accuracy, we also compare the computational complexity of different methods. As shown in Table VI, we list the number of parameters and inference time. The inference time is obtained by calculating how long it takes for each method to process a pair of bitemporal

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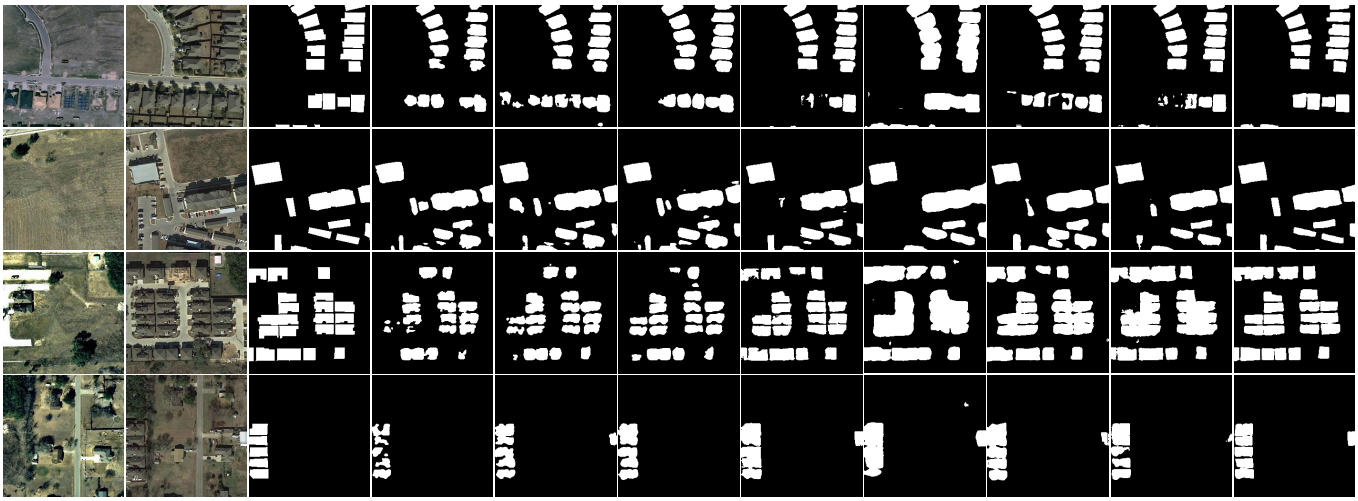


Fig. 7. Detection maps of comparison methods on LEVIR-CD. (a) Image at $t1$. (b) Image at $t2$. (c) Ground truth. (d) FC-EF [15]. (e) FC-Siam-conc [15]. (f) FC-Siam-diff [15]. (g) SNUNet [29]. (h) DSAMNet [28]. (i) BIT [33]. (j) ISNet [1]. (k) CANet.

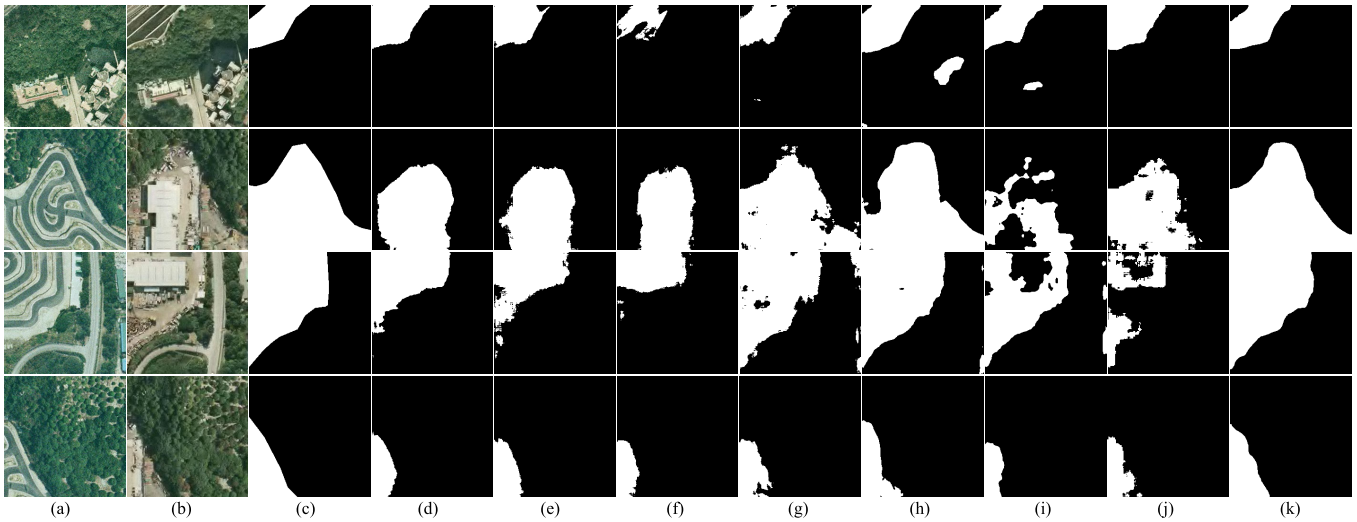


Fig. 8. Detection maps of comparison methods on SYSU-CD. (a) Image at $t1$. (b) Image at $t2$. (c) Ground truth. (d) FC-EF [15]. (e) FC-Siam-conc [15]. (f) FC-Siam-diff [15]. (g) SNUNet [29]. (h) DSAMNet [28]. (i) BIT [33]. (j) ISNet [1]. (k) CANet.

TABLE VI
COMPUTATIONAL COMPLEXITY COMPARISON OF CANET AND OTHER METHODS IN TERMS OF THE NUMBER OF PARAMETERS AND INFERENCE TIME

Model	FC-EF [15]	FC-Siam-conc [15]	FC-Siam-diff [15]	SNUNet [29]	DSAMNet [28]	BIT [33]	ISNet [1]	CANet
Parameters (M)	1.28	1.47	1.28	11.47	16.16	3.33	14.35	35.27
Inference time (ms)	38.86	39.93	41.00	46.75	44.99	52.69	-	68.74

images (256 × 256 pixels) with a GTX 3090 GPU. Note that, we do not report the inference time of ISNet because we cannot reimplement it by the public code, which is not compatible with GTX series GPUs. It can be observed that the computational complexity of FC-EF, FC-Siam-conc, and FC-Siam-diff is the lowest because they contain no extra complex modules. When introducing attention modules, the remaining comparison methods obviously increase the computational complexity. However, these modules indeed help improve the detection accuracy, as shown in Tables III–V. Among attention-based methods, ISNet outperforms the others in terms of F1, OA, and IoU within the acceptable range of the number of parameters. Compared with ISNet, our proposed CANet could further improve F1, OA, and IoU by at most 3.71%, 1.50%, and 5.10%, only less than 21M parameters are increased.

D. Ablation Study

1) Effectiveness of Each Module: Based on ResNet18, a total of four modules are proposed for remote sensing change detection. Hence, we report experimental results on the testing sets of three datasets to verify the performance of four modules. The experimental results are shown in

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Table VII. Here, TAM, CEM, CMM, and MFM in Table VII are the abbreviations for temporal attention module, context extraction module, context memory module, and multiscale fusion module, respectively. When any of these four modules is eliminated, we can observe that the detection performance would degrade to a certain extent. This phenomenon indicates that each of the proposed modules contributes to the detection result of CANet. It is worth noting that whether the context memory module is introduced or not seriously affects the performance of CANet. Specifically, IoU on three datasets will decrease by 3.23%, 1.41%, and 2.13%, respectively, if the context memory module is removed from CANet. We conjecture that pseudo-change problem alleviated by a context memory module is the main limitation on the detection performance. Furthermore, we visualize the detection results to qualitatively show whether the context memory module could deal with the pseudo-change problem. CANet without context memory module is denoted as *baseline*. As shown in Fig. 9, the marked areas in the first three rows illustrate the appearance difference between *vegetation* caused by seasonal change. Because *baseline* only compares intraimage context of two areas, it misclassifies these appearance differences as changes. When integrating the proposed context memory module into *baseline*, we can observe that the pseudo-change areas are perfectly eliminated. Even for slender objects in the fourth row, the context memory module could distinguish the real semantic change from the seasonal change. This visual result verifies the effectiveness of interimage context for solving the pseudo-change problem.

2) Analysis on Chosen Eigenvectors: Because the interimage context is inferred using the eigenvectors of the normalized Laplacian matrix, we conduct the ablation study on the number of chosen eigenvectors. Here, we select the optimal number from a candidate set $\{2, 4, 6, 8\}$. The experimental results are listed in Table VIII. It can be observed that IoU will first increase and then decrease as the number of chosen eigenvectors increases. When the number is set to 4, CANet obtains the best detection performance on three datasets. We speculate that dominating land-cover objects in the remote sensing scene could roughly be divided into four categories, including ground, building, water, and vegetation. When the number is too small, these eigenvectors are inadequate to distinguish different objects in a complex scene. On the other hand, the whole procedure of spectral clustering is based on an unsupervised manner, and thus, it is difficult to classify objects into too many fine-grained categories. Therefore, the detection performance degrades when the number is large. Besides, we illustrate the visualization results of eigenvectors in Fig. 10. As shown in Fig. 10, the first four eigenvectors could coarsely locate the object regions of specific semantic. Furthermore, we conduct the ablation study on the selected convolutional feature for spectral clustering. As shown in Table IX, we report the detection performance obtained by using the convolutional features from different blocks (Conv1, Conv2, Conv3, Conv4, and Conv5) of ResNet18 for spectral clustering. When the convolutional feature of Conv4 is selected for spectral clustering, the detection performance reaches the highest point. This is mainly because the convolutional feature of Conv4 contains appropriate spatial details and semantic information while others do not. As a result, spectral clustering could produce the regions with more sharp boundaries and precise semantic categories.

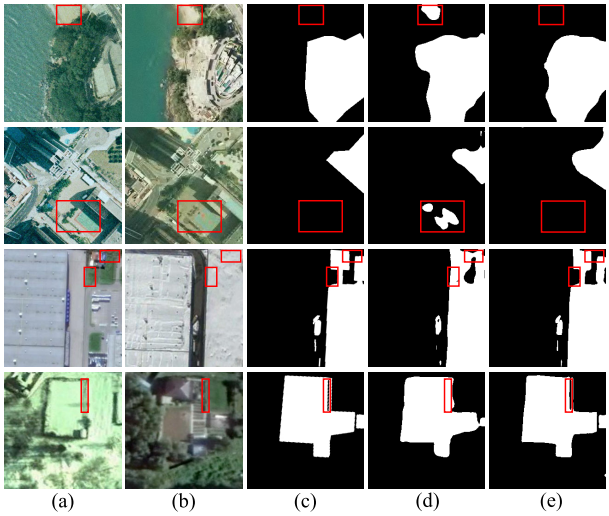


Fig. 9. Visual comparison between *baseline* and *baseline* with context memory module. (a) and (b) Bitemporal images. (c) Ground truth. (d) *Baseline*. (e) *Baseline* with context memory module.

TABLE VII
EFFECT OF FOUR PROPOSED MODULES ON THREE DATASETS

Modules				IoU		
TAM	CEM	CMM	MFM	SVCD	LEVIR-CD	SYSU-CD
✓	✓	✓	✓	**94.27**	**83.95**	**69.50**
✗	✓	✓	✓	92.45	83.43	69.00
✓	✓	✓	✗	93.49	83.57	69.07
✓	✗	✓	✓	92.04	83.06	68.33
✓	✓	✗	✓	91.04	82.54	67.37
✗	✗	✗	✗	90.48	81.92	66.06

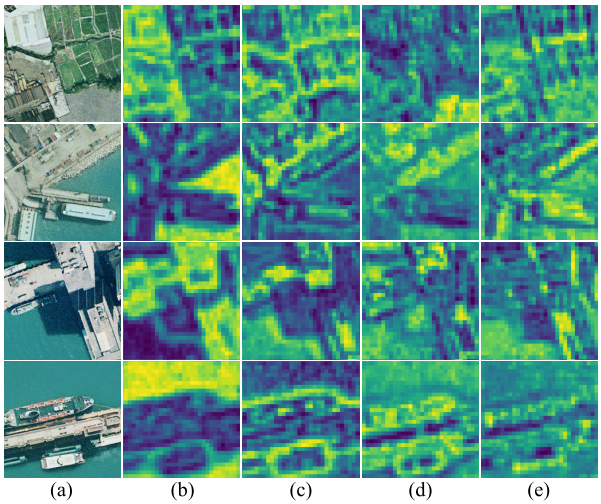


Fig. 10. Visualization results of eigenvectors. (a) Input image. (b) 1st eigenvector. (c) 2nd eigenvector. (d) 3rd eigenvector. (e) 4th eigenvector.

3) Effect of Different σ Values in Radial Basis Function: σ is the only hyperparameter of radial basis function used in spectral clustering. Here, we select the optimal σ from a candidate set $\{0.025, 0.25, 2.5, 25, 250\}$. Table X reports the

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TABLE VIII
EFFECT OF THE NUMBER OF CHOSEN EIGENVECTORS ON CHANGE DETECTION

Number	SVCD	LEVIR-CD	SYSU-CD
2	93.33	83.61	68.94
4	**94.27**	**83.95**	**69.50**
6	93.56	83.89	69.22
8	93.50	83.70	69.00

TABLE IX
EFFECT OF THE SELECTED CONVOLUTIONAL FEATURE FOR SPECTRAL CLUSTERING

Block	SVCD	LEVIR-CD	SYSU-CD
Conv1	93.03	83.17	67.83
Conv2	93.63	83.59	68.35
Conv3	93.97	83.88	68.95
Conv4	**94.27**	**83.95**	**69.50**
Conv5	93.90	83.83	68.51

TABLE X
EFFECT OF DIFFERENT σ VALUES ON THE DETECTION PERFORMANCE (IOU)

σ	SVCD	LEVIR-CD	SYSU-CD
0.025	93.84	82.98	68.84
0.25	94.00	83.23	69.25
2.5	**94.27**	**83.95**	**69.50**
25	93.95	83.47	69.03
250	93.74	83.02	68.79

effect of different σ values on the detection performance (IoU). When σ is set to 2.5, CANet obtains the highest IoU on three benchmark datasets, while IoU decreases as σ becomes larger or smaller. We conjecture that small σ increases the similarity between any two pixels even if they are from objects of different categories and vice versa. As a result, spectral clustering could not produce the accurate context representation of each underlying category in the scene, further leading to the degradation of the detection accuracy.

IV. CONCLUSION

In this article, a new method named CANet is proposed for remote sensing change detection. The problem CANet that tries to address is the pseudo change caused by external factors. To this end, context extraction and context memory modules are designed to simultaneously capture intraimage and interimage contexts, which are aggregated to provide more precise descriptions on changed objects of all categories. Besides, a temporal attention module is devised to extract the temporal dependency between bitemporal images and features from different scales are fully exploited for change inference via a multiscale fusion module. CANet achieves the best performance on three public datasets, in comparison with state-of-the-art methods. The ablation study on each proposed module is conducted and the results indicate that the performance of CANet heavily depends on the context memory module.

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